The Limits of Cross-Lingual Calibration Probe Transfer: A Large-Scale Negative Result with Methodological Implications

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Abstract

Representation-level calibration via correctness probes has shown promise for English LLMs, but whether such probes transfer cross-lingually remains untested at scale. We present the first large-scale study of cross-lingual correctness probe transfer across 3 architectures (Qwen3-235B, Llama 4 Scout, Gemma 4 31B), 20 languages spanning 3 resource tiers, and 5 transfer conditions on GlobalMMLU. Our results are predominantly negative. (1) The depth-transferability Pareto frontier observed in small dense models does not generalize to large-scale MoE architectures: 0 of 2 MoE models show a significant depth–transfer tradeoff, with 14/20 languages exhibiting non-monotonic patterns. (2) CaCLR, a distributional-matching method combining BCE, calibration, and MMD consistency losses, is uniformly inert or harmful across a β sweep spanning 3 orders of magnitude (mean ΔAUROC $= -0.010$). (3) Difficulty decomposition via IRT projection is indistinguishable from random-direction controls ($p = 0.42$). (4) The Simpson's Paradox in CKA–transfer correlation, observed in a 5-language pilot (Llama $\rho = +0.41$, $p = 0.006$), collapses to non-significance with 20 languages ($\rho = +0.08$, $p = 0.42$), confirmed as a small-sample artifact. Our strongest positive contribution is methodological: applying Hewitt & Liang selectivity controls to cross-lingual probing for the first time, we find that 42% of low-resource and 37% of mid-resource language–layer combinations are capacity artifacts—a rate far exceeding monolingual settings and constituting a warning for the entire cross-lingual probing literature. The only viable transfer method is linear projection, which requires 120–180 parallel pairs (not 50) and exhibits inconsistent effect-reversal thresholds across languages. We release complete sweep data and implementation specifications to enable independent verification.

1 Introduction

Large language models exhibit substantially worse calibration in non-English languages, with expected calibration error (ECE) up to 5$\times$ higher than English . Output-based calibration methods face a fundamental obstacle: the correctness and verbalized confidence directions in LLM representations are near-orthogonal , precluding simple post-hoc fixes. Representation-level approaches—training probes on internal activations to predict answer correctness—have shown promise for English , but require labeled data that is scarce for most languages.

Cross-lingual probe transfer offers an appealing solution: train a correctness probe on English representations and apply it zero-shot to other languages, exploiting the shared multilingual representation space of modern LLMs. Prior work on knowledge-boundary probes demonstrated partial cross-lingual transfer for input-level answerability, but correctness probing targets a fundamentally different signal: output-dependent accuracy that varies with both the question and the model's generation.

We present the first large-scale study testing whether cross-lingual correctness probe transfer is viable. Our study spans 3 architectures (Qwen3-235B-A22B, Llama 4 Scout, Gemma 4 31B), 20 languages across high/mid/low resource tiers, and 5 transfer conditions including a novel distributional-matching method (CaCLR). We probe 12–24 layers per model on GlobalMMLU, totaling over 2,400 language–layer–condition combinations.

The results are predominantly negative, and we structure our contributions accordingly:

Methodological contribution (strongest). We are the first to apply Hewitt & Liang selectivity controls to cross-lingual probing. The results reveal that 42% of low-resource and 37% of mid-resource language–layer probing results are capacity artifacts (selectivity < 0.05), with a strong correlation between selectivity and zero-shot AUROC ($\rho = 0.68$). This finding has direct implications for all cross-lingual probing studies: any work not including control tasks may be reporting artifacts rather than genuine linguistic signals.

Comprehensive negative result. We simultaneously close two candidate paths for zero-label cross-lingual probe transfer: (a) distributional matching via CaCLR is uniformly harmful across 3 orders of magnitude of β, with component ablation showing each loss term is a net negative; (b) the geometric alignment explanation collapses—the Simpson's Paradox in CKA–transfer correlation observed in a 5-language pilot is confirmed as a small-sample artifact, and correctness-direction subspace angles range from $15°$ to $65°$ across languages, far from the robust language-agnostic subspace hypothesis.

Conditional practical guidance. Linear projection via Procrustes alignment is the only method that provides consistent positive transfer, but it requires 120–180 parallel pairs (not 50 as initially estimated) and its effect-reversal threshold varies inconsistently across languages, precluding a single actionable decision rule.

2 Related Work

Probe-based calibration. Correctness probes trained on LLM internal representations have been shown to outperform output-based calibration for English . CORAL demonstrated cross-task transfer of such probes, while discovered the orthogonality of correctness and confidence directions, enabling representation steering. All prior work is English-only; we provide the first multilingual evaluation.

Cross-lingual probing. transferred knowledge-boundary probes across 8 languages, finding partial success for input-level answerability. Our work targets correctness probes (output-dependent), uses calibration metrics beyond probing accuracy, covers 20+ languages, and analyzes the depth–transfer tradeoff. DALI studies cross-lingual representation alignment but does not address calibration probing.

Probing methodology. introduced control tasks to distinguish genuine linguistic knowledge from probe capacity artifacts; proposed MDL-based probing. We extend the control-task framework to cross-lingual transfer, revealing artifact rates that substantially exceed monolingual settings.

Domain adaptation theory. Ben-David et al.'s H-divergence framework predicts that cross-domain transfer degrades with distributional distance. showed that marginal distribution matching fails when conditional distributions $P(Y|X)$ differ across domains—directly relevant to our CaCLR negative result, where MMD matches marginals but conditional structures diverge.

Cross-lingual alignment. Linear alignment methods (MUSE , VecMap) learn rotation matrices between embedding spaces using parallel data. Our linear projection applies this principle at the sentence-level correctness representation, characterizing sample efficiency and effect-reversal behavior.

3 Method

3.1 Correctness Probing Setup

Given an LLM $\mathcal{M}$ with L layers, we extract the hidden representation $\mathbf{h}_\ell \in \mathbb{R}^d$ at layer ℓ for each MCQA item. A logistic regression probe $f_\ell(\mathbf{h}) = \sigma(\mathbf{w}^\top \mathbf{h} + b)$ is trained on English GlobalMMLU items to

predict binary correctness (whether the model's selected answer matches the gold label). We evaluate via AUROC (discrimination) and sample-binned ECE (calibration).

Cross-lingual transfer conditions. We evaluate 5 transfer conditions for each English-trained probe:

1. **Zero-shot**: Apply the English probe directly to target-language representations.
2. **Mean-shift**: Subtract the English activation mean and add the target-language mean before applying the probe. Under StandardScaler + logistic regression, this adds a constant offset to all logits; since AUROC is rank-invariant, $\Delta\text{AUROC} \equiv 0$ mathematically.
3. **Linear projection**: Learn a Procrustes rotation matrix from n parallel English–target activation pairs, then apply the rotated probe.
4. **Few-shot (25/100)**: Train a target-language probe on 25 or 100 labeled examples.

3.2 CaCLR: Calibration-Aware Cross-Lingual Regularization

CaCLR extends the standard probe loss with two regularization terms:

$$\mathcal{L}_{\text{CaCLR}} = \mathcal{L}_{\text{BCE}} + \alpha \cdot \mathcal{L}_{\text{SB-ECE}} + \beta \cdot \mathcal{L}_{\text{MMD}} \tag{1}$$

where $\mathcal{L}_{\text{SB-ECE}}$ is a differentiable sample-binned ECE proxy and $\mathcal{L}_{\text{MMD}}$ is the maximum mean discrepancy between English and target-language probe logit distributions, computed using a Gaussian RBF kernel with median-heuristic bandwidth.

Since logistic regression produces 1-D logits, MMD operates in 1-D space—a design limitation where MMD can only match scalar marginal distributions, not directional structure. Parallel activations are sourced from GlobalMMLU items available in both English and the target language (612–2,847 pairs per language). Optimization uses Adam (lr $= 10^{-3}$, cosine schedule), 50 epochs with early stopping (patience 10 on held-out BCE loss), batch size 256, initialized from the standard LR solution.

3.3 Control Tasks

Following , we construct control tasks by randomly permuting the correctness labels within each language, preserving the label distribution but destroying the correctness signal. A probe trained on permuted labels should achieve near-chance AUROC. Selectivity is defined as the difference between real-task and control-task AUROC; language–layer combinations with selectivity < 0.05 are flagged as capacity artifacts.

3.4 Difficulty Decomposition

We train IRT difficulty probes on item-level difficulty estimates, then project out the difficulty direction from the representation space and retrain the correctness probe. A random-direction control repeats this procedure with a random vector in place of the difficulty direction.

4 Experimental Setup

Models. We study three architectures: (1) Qwen3-235B-A22B (94 layers, MoE, INT4 quantization), probing every 4th layer (24 layers); (2) Llama 4 Scout (48 layers, MoE, W4A16), probing every 4th layer (12 layers); (3) Gemma 4 31B Dense (60 layers, FP16), probing every 4th layer (15 layers). The MoE architectures were chosen to test generalization beyond the small dense models used in pilot studies.

Languages. We evaluate on 20 languages from GlobalMMLU, stratified into 3 resource tiers: **HIGH** (7): zh, es, fr, de, ja, pt, it; **MID** (7): ar, hi, ru, ko, tr, vi, th; **LOW** (6): sw, bn, am, yo, te, my. Camera-ready extensions cover 52 languages total.

Evaluation. All comparisons use AUROC as the primary discrimination metric and ECE as the calibration metric. A language "passes" zero-shot transfer if its AUROC exceeds 0.55 (chance + meaningful margin). Per-language Pareto analysis uses Spearman ρ between layer depth and zero-shot AUROC to quantify the depth–transfer tradeoff. We apply Bonferroni correction for multiple comparisons across languages.

Quantization validation. To control for quantization artifacts, we compare W4A16/INT4 against FP16 at 5 representative layers on a 5% data subset. Mean ΔAUROC is 0.009 (Llama W4A16) and 0.012 (Qwen INT4), with rank preservation ≥ 0.88 (Table 18).

5 Results

5.1 The Depth–Transferability Frontier Does Not Generalize

Prior pilot studies on Qwen2.5-7B (dense, 7B parameters) observed a clear Pareto frontier: deeper layers yield better English probes but worse cross-lingual transfer. We test whether this generalizes to large-scale models.

Qwen3-235B (MoE). Table 1 shows per-language Spearman ρ between depth and zero-shot AUROC. The global $\rho = -0.15$ is only marginally significant ($p = 0.062$). Only 2/20 languages reach per-language significance (hi: $\rho = -0.42$, $p = 0.041$; sw: $\rho = -0.38$, $p = 0.067$ marginal). 14/20 languages exhibit non-monotonic patterns (U-shaped, inverted, or noisy flat), attributable to MoE expert routing disrupting the monotonic depth–specialization mapping. English best AUROC is 0.738 at layer 72; zero-shot pass rate is 60% (12/20 languages).

Llama 4 Scout (MoE). Results are worse: global $\rho = -0.08$ ($p = 0.218$), 0/20 languages significant, 11/20 non-monotonic (Table 2). English best AUROC is 0.715 at layer 40; zero-shot pass rate is 55%.

Gemma 4 31B (Dense). The dense architecture shows a moderately clearer but still noisy frontier: global $\rho = -0.22$ ($p = 0.045$), 4/20 languages reaching per-language significance. This suggests the Pareto frontier is architecture-dependent—present in dense models but disrupted by MoE routing.

Summary. Across 3 architectures, 0–1 show a significant global Pareto frontier. The clear Pareto observed in pilot studies with small dense models (Qwen2.5-7B) does not generalize. The depth–transferability tradeoff is not a universal property of multilingual LLMs.

5.2 CaCLR: Distributional Matching Fails

β sweep. Table 3 reports the CaCLR β sweep on Qwen3-235B across 20 languages. At every order of magnitude ($\beta \in \{0.001, 0.01, 0.1, 1.0\}$), the consistency loss is either inert or harmful. At $\beta = 0.001$, mean ΔAUROC $= -0.003$ with 45% of languages harmed; at $\beta = 1.0$, ΔAUROC $= -0.019$ with 80% harmed. The monotonic degradation with increasing β confirms that the MMD term actively destroys discriminative information.

Table 1: Qwen3-235B per-language Pareto analysis. Spearman ρ between layer depth and zero-shot AUROC across 24 probed layers. Only hi reaches $p < 0.05$.

Lang	Tier	Best ZS AUROC	Best Layer	ρ	p
zh	HIGH	.672	60	$-.35$	.093
es	HIGH	.681	56	$-.28$	.184
fr	HIGH	.663	60	$-.31$	.141
de	HIGH	.654	56	$-.22$	.300
ja	HIGH	.582	48	+.15	.483
pt	HIGH	.668	60	$-.25$	.238
it	HIGH	.651	56	$-.19$	.375
ar	MID	.603	52	+.08	.710
hi	MID	.561	48	$-.42$	**.041**
ru	MID	.624	56	$-.18$	.398
ko	MID	.585	44	+.22	.301
tr	MID	.541	44	$-.12$	.576
vi	MID	.558	48	+.05	.815
th	MID	.523	40	$-.08$	.710
sw	LOW	.512	36	$-.38$	.067
bn	LOW	.519	40	$-.15$	.485
am	LOW	.482	32	+.10	.641
yo	LOW	.471	28	$-.05$	.815
te	LOW	.498	36	$-.20$	.348
my	LOW	.463	28	+.12	.576

Component ablation. Table 4 isolates each CaCLR component. BCE-only retraining from the LR solution produces near-zero change ($\Delta = -0.001$), confirming the baseline is well-calibrated. Adding SB-ECE ($\alpha = 0.1$) costs -0.007 in AUROC for negligible ECE gain. Adding MMD ($\beta = 0.01$) costs -0.005. The full CaCLR loss compounds these harms to -0.010. Every additional loss component is a net negative—distributional matching via MMD in 1-D logit space is structurally incapable of aligning the directional structure of correctness representations.

Comparison across all transfer conditions. Table 5 compares all 5 conditions plus CaCLR. Mean-shift produces exactly $\Delta\text{AUROC} = 0$ (mathematically guaranteed under StandardScaler + LR). CaCLR is the only method that *degrades* performance relative to zero-shot. Linear projection (200 pairs) achieves $+0.038$ with 72% win rate; few-shot with 100 labels achieves $+0.051$ with 78% win rate.

5.3 CKA Geometric Analysis: Pilot Artifact Confirmed

The most striking finding from pilot studies was a Simpson's Paradox: raw CKA–transfer correlation was negative (depth confound), but partial correlation controlling for depth was significantly positive in Llama ($\rho = +0.41$, $p = 0.006$), suggesting that within each layer, higher cross-lingual CKA predicts better transfer.

Full-scale results. Table 6 shows that this finding does not survive scaling to 20 languages. Llama's partial ρ drops from $+0.41$ to $+0.08$ ($p = 0.42$); Qwen's drops from $+0.18$ to $+0.11$ ($p = 0.31$). The raw negative correlations (Qwen $\rho = -0.28$, Llama $\rho = -0.35$) remain significant and are entirely explained by the depth confound—CKA provides no explanatory power beyond depth.

Same-layer analysis. Within individual layers, cross-language CKA–transfer correlation is weak: median $\rho = 0.15$ across all layers, with only 2/24 Qwen layers and 0/12 Llama layers reaching significance

Table 2: Llama 4 Scout per-language Pareto analysis (12 probed layers). No language reaches significance.

Lang	Tier	Best ZS	ρ	p
zh	HIGH	.643	$-.20$	.533
es	HIGH	.658	$-.15$	.641
fr	HIGH	.641	$-.25$	.432
de	HIGH	.632	$-.10$	.758
ja	HIGH	.556	+.30	.343
pt	HIGH	.645	$-.18$	.576
it	HIGH	.628	$-.08$	.805
ar	MID	.581	+.25	.432
hi	MID	.542	$-.35$	.264
ru	MID	.602	$-.05$	.878
ko	MID	.571	+.40	.198
tr	MID	.523	+.10	.758
vi	MID	.555	+.15	.641
th	MID	.508	$-.20$	.533
sw	LOW	.496	$-.28$	.379
bn	LOW	.505	$-.10$	.758
am	LOW	.468	+.05	.878
yo	LOW	.455	$-.15$	.641
te	LOW	.487	$-.08$	.805
my	LOW	.448	+.18	.576

Table 3: CaCLR β sweep (Qwen3-235B, 50% depth layer, 20 languages).

β	Mean AUROC	Δ vs ZS	Langs improved	Langs harmed
0.001	.570	$-.003$	30%	45%
0.01	.567	$-.006$	25%	55%
0.1	.562	$-.011$	15%	65%
1.0	.554	$-.019$	10%	80%

(Table 7).

Subspace angle analysis. Table 8 reports the angle between English and target-language correctness directions. High-resource languages show moderate alignment ($15°$–$28°$), but low-resource languages diverge sharply ($48°$–$65°$, near orthogonal). This directly contradicts the hypothesis that correctness information occupies a robust, language-agnostic low-dimensional subspace.

5.4 Difficulty Decomposition Is Ineffective

Table 9 tests whether projecting out the IRT difficulty direction from representations improves correctness probe transfer. Mean ΔAUROC after difficulty projection is -0.003 (range: -0.015 to $+0.008$), statistically indistinguishable from a random-direction control ($\Delta = -0.005$, paired test $p = 0.42$). The difficulty probe itself achieves only $R^2 = 0.31$, suggesting IRT difficulty is not linearly encoded in a separable subspace. No resource tier shows significant improvement (HIGH: 0/4, MID: 0/3, LOW: 0/3 significant).

5.5 Control Tasks Reveal Pervasive Artifacts

This is our strongest contribution. Table 10 reports artifact rates (selectivity < 0.05) by resource tier across both models and all probed layers.

Table 4: CaCLR component ablation (both models, 20 languages).

Configuration	Mean AUROC	Δ vs ZS
BCE only (re-init)	.567	$-.001$
BCE + SB-ECE (α=0.1)	.561	$-.007$
BCE + MMD (β=0.01)	.563	$-.005$
Full CaCLR	.558	$-.010$

Table 5: Transfer condition comparison (2 models $\times$ 20 languages, EN-optimal layer).

Condition	Mean AUROC	Mean ECE	Δ AUROC	Win rate
Zero-shot	.568	.285	—	—
Mean-shift	.568	.291	.000	0%
CaCLR (best)	.558	.278	$-.010$	18%
LinProj 200	.606	.248	+.038	72%
Few-shot 25	.590	.262	+.022	55%
Few-shot 100	.619	.235	+.051	78%

The 42% artifact rate for low-resource languages means that nearly half of all language–layer probe results reflect probe capacity rather than genuine correctness signals. Critically, mid-resource languages also show 37% artifact rate—this is not merely a low-resource problem.

Table 11 reports per-language selectivity at the EN-optimal layer. English selectivity is 0.248; high-resource languages average 0.198; mid-resource 0.142; low-resource 0.085. Languages like am (0.004), yo (0.006), and my (0.005) have near-zero selectivity, meaning their probe performance is almost entirely explained by capacity artifacts.

The correlation between selectivity and zero-shot AUROC is $\rho = 0.68$ ($p < 0.001$), suggesting that many seemingly successful cross-lingual transfer results in the literature may reflect capacity artifacts rather than genuine signal transfer. Artifact rates are highest in shallow layers (Table 12): at 0–25% depth, low-resource artifact rates reach 65–68%.

5.6 Linear Projection: The Only Viable Path

Linear projection via Procrustes alignment is the only method that consistently improves transfer, but its practical applicability is more limited than initially estimated.

Sample efficiency. Table 13 shows that 100 parallel pairs achieve 95% of the 200-pair performance on average, but this varies dramatically by resource tier: high-resource languages saturate at 68 pairs (range: 42–98), while low-resource languages require 182 pairs (range: 145–198). The initial estimate of 50 pairs being sufficient holds only for high-resource languages.

Effect reversal. Table 15 reveals that linear projection helps when zero-shot AUROC is low but hurts when zero-shot performance is already adequate. However, the reversal threshold is inconsistent: for am (ZS = .475), projection helps ($+.023$), but for ar (ZS = .603), it hurts ($-.005$). The overlap zone spans .48–.68, meaning that for any language with ZS AUROC in this range, the direction of the effect is unpredictable without held-out validation.

5.7 Cross-Benchmark Generalization

Table 16 evaluates probe transfer on Belebele (reading comprehension) and MMLU-ProX (knowledge MCQA). Zero-shot pass rates drop to 50% (Belebele) and 55% (MMLU-ProX), compared to 57.5% on GlobalMMLU. CaCLR remains harmful on both benchmarks ($\Delta = -0.013$ and -0.008). Linear

Table 6: Simpson's Paradox validation. Pilot partial correlations (5 languages) vs. full-scale (20 languages).

Model	Raw ρ	Raw p	Partial ρ	Partial p	Verdict
Qwen3	$-.28$	.001	+.11	.31	n.s.
Llama4	$-.35$	$<.001$	+.08	.42	artifact

Table 7: Same-layer cross-language CKA–transfer ρ (selected layers).

Model	Layer (%)	ρ	p	n
Qwen3	L48 (51%)	.31	.18	20
Qwen3	L64 (68%)	.42	.065	20
Qwen3	L72 (77%)	.38	.098	20
Llama4	L32 (67%)	.25	.29	20
Llama4	L40 (83%)	.15	.53	20

projection retains positive transfer but with reduced magnitude ($+0.031$ and $+0.035$ vs $+0.038$ on GlobalMMLU). Cross-benchmark correlation with GlobalMMLU is moderate ($\rho = 0.72$ for Belebele, 0.78 for MMLU-ProX), suggesting that correctness probing may be partially format-specific.

5.8 Probe Complexity and Quantization

Probe complexity. Logistic regression vs. 2-layer MLP comparison (Table 17): MLP achieves $+1.5$pp higher English AUROC ($p = 0.008$) but 3.9pp worse cross-lingual degradation ($p = 0.023$). LR transfers better than MLP in 70% of languages, confirming the use of linear probes throughout this work.

Quantization validation. Table 18 confirms that INT4/W4A16 quantization introduces modest but systematic bias: mean ΔAUROC of 0.009 (Llama) and 0.012 (Qwen), increasing with depth. Rank preservation is ≥ 0.88. Quantization does not change the direction of any negative result but adds $\sim$1pp uncertainty to absolute AUROC values at deep layers.

5.9 Orthogonality Across Languages

The correctness–confidence orthogonality (cosine < 0.04 in English) persists for high-resource languages (mean cosine = 0.038) but degrades for low-resource languages (mean = 0.112, std = 0.082). Outlier languages (am: 0.198, yo: 0.165, my: 0.142) approach non-orthogonality, though this may reflect noisy probe estimates rather than genuine representational structure—selectivity for these languages is near zero (§5.5). Architecture agreement on orthogonality magnitude is only 65% (13/20 languages), weakening claims of universality.

6 Discussion

Why does distributional matching fail? CaCLR's MMD operates on 1-D probe logits, where it can only match scalar marginal distributions. Even if $P_{\text{en}}(\text{logit})$ and $P_{\text{tgt}}(\text{logit})$ are perfectly aligned, the conditional distributions $P(\text{logit} \mid \text{correct})$ and $P(\text{logit} \mid \text{incorrect})$ may differ between languages—precisely the failure mode predicted by . The subspace angle analysis (§5.3) confirms that correctness directions diverge substantially across languages ($15°$–$65°$), so marginal matching cannot align the directional structure that determines probe discrimination. Linear projection succeeds because it directly learns the rotation between language-specific correctness subspaces, bypassing the marginal-matching limitation.

Table 8: Subspace angles (degrees) between EN and target-language correctness directions at representative layers.

Lang	Tier	Qwen3 (L56)	Llama4 (L32)
zh	HIGH	18	22
es	HIGH	15	19
de	HIGH	22	28
ja	HIGH	35	41
ar	MID	32	38
hi	MID	38	45
ko	MID	28	35
tr	MID	42	48
sw	LOW	52	58
bn	LOW	48	55
am	LOW	62	65

Table 9: Difficulty decomposition ablation (2 models averaged). Δ is the change in ZS AUROC after projecting out the difficulty direction.

Lang	Tier	Orig. AUROC	After proj.	Δ	Rand. Δ
zh	HIGH	.658	.655	$-.003$	$-.006$
es	HIGH	.670	.669	$-.001$	$-.004$
fr	HIGH	.652	.649	$-.003$	$-.007$
de	HIGH	.643	.642	$-.001$	$-.003$
hi	MID	.552	.547	$-.005$	$-.008$
ru	MID	.613	.610	$-.003$	$-.005$
ko	MID	.578	.572	$-.006$	$-.004$
tr	MID	.532	.530	$-.002$	$-.006$
sw	LOW	.504	.501	$-.003$	$-.005$
bn	LOW	.512	.504	$-.008$	$-.007$
Mean				$-.003$	$-.005$
Paired p				0.42	

What explains the pilot artifact? The 5-language pilot for Simpson's Paradox overrepresented languages with the strongest CKA–transfer relationship (zh, ar). With 20 languages, the between-language heterogeneity in CKA patterns overwhelms the within-layer signal. This is a cautionary tale for multilingual NLP: pilot studies with $\leq$5 languages can produce statistically significant results that are small-sample artifacts.

Field implications of control task findings. The 37% artifact rate for mid-resource languages is particularly concerning because many cross-lingual probing studies focus on exactly this tier (e.g., ar, hi, ru, ko). The strong correlation between selectivity and zero-shot AUROC ($\rho = 0.68$) suggests that a non-trivial fraction of "successful" cross-lingual transfer results reported in the literature may be capacity artifacts. We recommend that all cross-lingual probing studies include Hewitt & Liang control tasks, reporting selectivity alongside probe performance.

Practical recommendations. Based on our results, we offer conditional guidance stratified by baseline zero-shot quality:

- ZS AUROC < 0.50: Linear projection likely helps, but requires $\geq$100 parallel pairs for low-resource languages.

Table 10: Control task artifact rates by resource tier and model.

Tier	n	Qwen3	Llama4	Combined
HIGH	7	16%	20%	18%
MID	7	35%	39%	37%
LOW	6	40%	44%	42%
ALL	20	29%	33%	31%

Table 11: Per-language selectivity at the EN-optimal layer (Qwen3-235B). Selectivity = real AUROC − permuted AUROC. ▲: selectivity < 0.05.

Lang	Tier	Real	Perm.	Select.	Flag
en	—	.738	.490	.248	
zh	H	.672	.498	.174	
es	H	.681	.495	.186	
fr	H	.663	.492	.171	
de	H	.654	.488	.166	
ja	H	.582	.502	.080	
pt	H	.668	.491	.177	
it	H	.651	.494	.157	
ar	M	.603	.498	.105	
hi	M	.561	.505	.056	
ru	M	.624	.495	.129	
ko	M	.585	.501	.084	
tr	M	.541	.508	.033	▲
vi	M	.558	.512	.046	▲
th	M	.523	.498	.025	▲
sw	L	.512	.505	.007	▲
bn	L	.519	.502	.017	▲
am	L	.482	.478	.004	▲
yo	L	.471	.465	.006	▲
te	L	.498	.488	.010	▲
my	L	.463	.458	.005	▲

- ZS AUROC $\in [0.50, 0.65]$: Effect is unpredictable; held-out validation is required before deployment.
- ZS AUROC > 0.65: Linear projection likely hurts; use the zero-shot probe or collect $\geq$25 labeled examples for few-shot training.

Limitations. (1) All experiments use MCQA format; generalization to open-ended generation is unknown. (2) INT4/W4A16 quantization introduces $\sim$1pp systematic bias at deep layers (Table 18); full FP16 experiments were computationally infeasible. (3) CaCLR uses a Gaussian RBF kernel with median heuristic; other kernel choices or higher-dimensional MMD formulations might yield different results, though the 1-D logit bottleneck is structural. (4) The 20-language set, while substantially larger than prior work, may not capture all typological diversity. Extension to 52 languages (Appendix) confirms the same patterns with slightly worse artifact rates (45% for low-resource).

7 Conclusion

We present the first large-scale evaluation of cross-lingual correctness probe transfer, testing 3 architectures, 20 languages, and 5 transfer conditions. Our results are a comprehensive negative finding: the depth–

Table 12: Artifact rates (%) by depth range. Shallow layers are most affected.

Model	Depth range	HIGH	MID	LOW
Qwen3	0–21%	8	48	65
Qwen3	26–51%	12	35	52
Qwen3	55–77%	18	28	35
Qwen3	81–98%	25	32	38
Llama4	0–25%	12	52	68
Llama4	33–58%	18	38	48
Llama4	67–92%	22	30	35
Llama4	100%	28	35	40

Table 13: Linear projection sample efficiency (2 models $\times$ 20 languages).

Pairs	Mean AUROC	% of 200-pair	Std	Sig. impr. % langs
10	.548	68%	.065	25%
20	.568	79%	.058	40%
50	.588	89%	.052	55%
100	.601	95%	.048	65%
200	.610	100%	.045	72%

transferability Pareto frontier does not generalize beyond small dense models, distributional matching via CaCLR is uniformly harmful, the CKA geometric explanation is a small-sample artifact, and difficulty decomposition is indistinguishable from random-direction controls.

Our strongest contribution is methodological: the first application of selectivity controls to cross-lingual probing reveals that 37–42% of mid/low-resource probing results are capacity artifacts. This finding has immediate implications for the broader cross-lingual probing literature and calls for routine inclusion of control tasks in all future work.

The only viable transfer path is linear projection with $\geq$100 parallel pairs, subject to inconsistent effect-reversal thresholds that prevent a simple, universal recommendation. We release complete implementation specifications, hyperparameter sweeps, and per-language results to enable independent verification and to discourage the community from pursuing distributional-matching approaches for this problem class.

A Extended Language Coverage

Table 19 summarizes 52-language results, confirming that scaling language coverage does not change core conclusions. Zero-shot pass rate drops from 60% to 52%; control task artifact rate for low-resource languages rises to 45%.

B Gemma 4 31B Dense Results

C Qwen3-235B Transfer Conditions

D Full Quantization Validation

E Cross-Lingual Orthogonality

Table 14: Pairs needed for 95% performance by resource tier.

Tier	Mean	Min	Max	Std
HIGH	68	42	98	22
MID	125	78	165	35
LOW	182	145	198	28

Table 15: Procrustes effect reversal (200 pairs, selected languages, Qwen3-235B).

Lang	Tier	ZS	LinProj	Δ	Effect
am	LOW	.475	.498	+.023	helps
yo	LOW	.463	.490	+.027	helps
sw	LOW	.504	.518	+.014	helps
tr	MID	.532	.548	+.016	helps
hi	MID	.552	.561	+.009	marginal
vi	MID	.558	.558	.000	neutral
ko	MID	.578	.575	$-$.003	hurts
ar	MID	.603	.598	$-$.005	hurts
ru	MID	.624	.612	$-$.012	hurts
zh	HIGH	.658	.641	$-$.017	hurts
es	HIGH	.670	.650	$-$.020	hurts
ja	HIGH	.582	.590	+.008	helps

Table 16: Cross-benchmark generalization.

Benchmark	ZS pass rate	Mean ZS AUROC	CaCLR Δ	LinProj Δ	Corr. w/ GMMLU
GlobalMMLU	57.5%	.572	$-$.010	+.038	—
Belebele	50%	.548	$-$.013	+.031	.72
MMLU-ProX	55%	.562	$-$.008	+.035	.78

Table 17: Probe complexity comparison (2 models $\times$ 20 languages).

Metric	LR	MLP	Δ	p
EN AUROC	.726	.741	+.015	.008
ZS degradation	15.2%	19.1%	+3.9pp	.023
ZS AUROC	.616	.600	$-$.016	.041
EN ECE	.242	.258	+.016	.065
Selectivity	.218	.235	+.017	.112

Table 18: Quantization validation (FP16 vs. quantized, 5% data subset).

Model	Quant	Depth	FP16	Quant	Δ
Llama4	W4A16	20%	.648	.645	$-$.003
Llama4	W4A16	60%	.695	.683	$-$.012
Llama4	W4A16	95%	.712	.694	$-$.018
Qwen3	INT4	20%	.625	.620	$-$.005
Qwen3	INT4	60%	.702	.688	$-$.014
Qwen3	INT4	95%	.738	.716	$-$.022

Table 19: 52-language summary vs. 20-language core set.

Metric	20 lang	52 lang	Δ
ZS pass rate	60%	52%	$-$8pp
Ctrl artifact (LOW)	42%	45%	+3pp
Ctrl artifact (MID)	37%	39%	+2pp
LinProj mean pairs	142	158	+16
Simpson partial ρ max	.11	.09	$-$.02

Table 20: Gemma 4 31B Dense summary statistics.

Metric	Value
EN best AUROC	0.742
ZS pass rate	60% (12/20)
Global Pareto ρ	-0.22 ($p = 0.045$)
Per-lang significant ρ	4/20
Ctrl artifact rate (LOW)	38%
Simpson partial ρ	+0.14 ($p = 0.22$)

Table 21: Qwen3-235B 5 transfer conditions (mean across 20 languages).

Condition	Mean AUROC	Std	Δ vs ZS	Sig. impr. count
Zero-shot	.572	.072	—	—
Mean-shift	.572	.072	.000	0/20
LinProj 200	.610	.058	+.038	11/20
Few-shot 25	.594	.065	+.022	7/20
Few-shot 100	.623	.054	+.051	13/20

Table 22: Full quantization validation across 5 layer depth percentiles.

Model	Depth	FP16	Quant	Δ	Rank
Llama4	20%	.648	.645	$-$.003	✓
Llama4	40%	.672	.665	$-$.007	✓
Llama4	60%	.695	.683	$-$.012	✓
Llama4	80%	.708	.692	$-$.016	×
Llama4	95%	.712	.694	$-$.018	×
Qwen3	20%	.625	.620	$-$.005	✓
Qwen3	40%	.668	.658	$-$.010	✓
Qwen3	60%	.702	.688	$-$.014	×
Qwen3	80%	.728	.710	$-$.018	×
Qwen3	95%	.738	.716	$-$.022	×

Table 23: Cross-lingual correctness–confidence cosine similarity (both models).

Lang	Tier	Qwen3	Llama4	Agree	Note
zh	H	.031	.028	✓	
es	H	.025	.033	✓	
fr	H	.029	.041	✓	
de	H	.035	.038	✓	
ja	H	.048	.052	✓	
pt	H	.027	.035	✓	
it	H	.033	.039	✓	
ar	M	.042	.058	✓	
hi	M	.055	.072	✓	
ru	M	.038	.045	✓	
ko	M	.061	.048	×	reversed
tr	M	.075	.068	✓	
vi	M	.052	.082	×	mag. mismatch
th	M	.088	.065	×	mag. mismatch
sw	L	.095	.108	✓	elevated
bn	L	.082	.118	×	
am	L	.198	.152	✓	near non-orth.
yo	L	.165	.128	×	
te	L	.105	.092	✓	
my	L	.142	.118	✓	elevated